1. The basic circuit

The simplest circuit is a battery, some connecting wire and a resistor in a loop. The battery creates a potential difference (voltage) that pushes electrons through the wire. Current flows out of the positive terminal, through the resistor, and back to the battery. Conventional current is drawn from positive to negative, though electrons drift the other way. Every practical circuit in this course - meters, rheostats, loads - builds on this single loop, so master the loop before adding instruments. MARKER-CIRCUIT-P1.

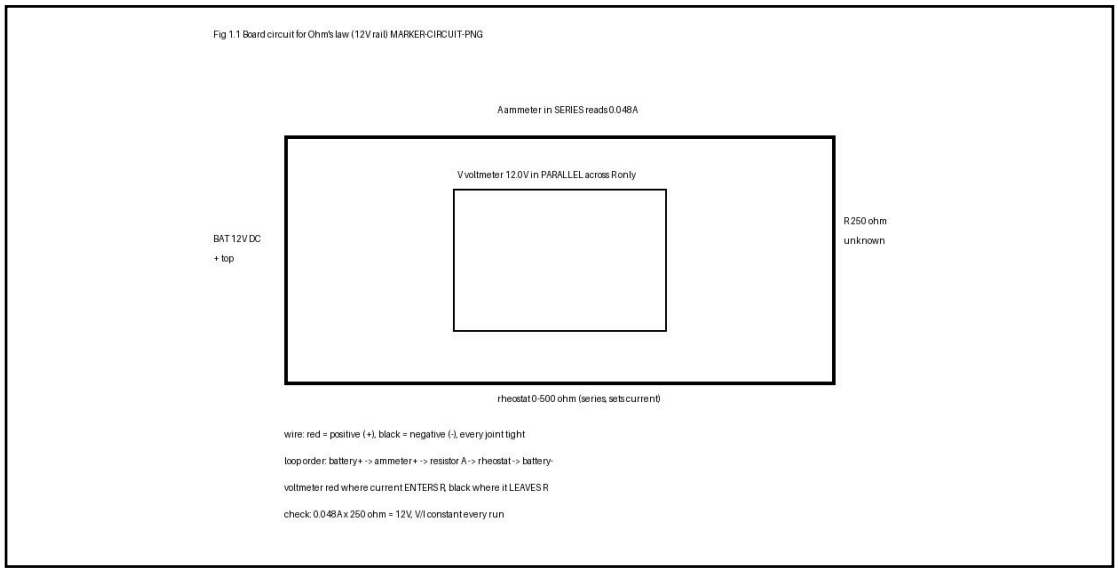


Fig 1.1: battery-ammeter-resistor-rheostat loop, voltmeter in parallel.

2. The water-hose analogy

Think of a garden water hose. Battery voltage is the water-supply pressure: higher pressure pushes harder. The outlet valve is the resistor: closing it reduces the flow. The water flow itself is the electric current. Increase resistance and flow drops; open the valve (less resistance) and flow rises. A kinked hose is a large resistor, a wide clean pipe is a small one, and the tap you open is exactly what the rheostat does in the electric loop. MARKER-HOSE-P2.

3. Ohm's law and the triangle

Ohm's law links the three quantities: V = I x R. Draw it as a triangle with V on top and I, R below. Cover the value you want: cover V and you read I multiplied by R; cover I and you read V divided by R; cover R and you read V divided by I. The triangle is a memory aid, not physics: the physics is that current through a fixed resistor is directly proportional to the voltage across it, doubling V doubles I. It holds for ohmic conductors at constant temperature. MARKER-TRIANGLE-P3.

4. Units and symbols

In engineering voltage is denoted V and measured in volts. Resistance is denoted R (or the omega sign) and measured in ohms. Current is denoted I and measured in amps. So V=IR reads volts = amps x ohms. Milliamps are thousandths of an amp: 0.048 A is 48 mA, 0.024 A is 24 mA. Always convert milliamps to amps before multiplying by ohms, otherwise the answer is off by a factor of a thousand. MARKER-UNITS-P4.

5. Worked example: 12V across 250 ohms

Take a 12 volt battery with a 250 ohm resistor across it. Cover I in the triangle: I = V / R. 12 volts divided by 250 ohms gives 0.048 amps, which is 48 milliamps - all without using a meter. Connecting wire has a small resistance too, but here the total is treated as 250 ohms. Check by back-substitution: 0.048 A x 250 ohm = 12 V, which matches the battery, so the loop is consistent. MARKER-EXAMPLE-P5.

Measured V-I pairs for R = 250 ohm (excerpt):

voltage_V	current_mA	resistance_ohm
3.0	12.0	250
6.0	24.0	250
9.0	36.0	250
12.0	48.0	250

6. Series, parallel and the board practical

An ammeter joins the loop in series so the full current passes through it. A voltmeter connects in parallel across the resistor only: red (positive) where current enters, black (negative) where it leaves. A rheostat in series varies the current for the experiment. Red terminals are positive, black are negative, and every connection must be tight. In series the same current flows everywhere; in parallel the same voltage appears across each branch, which is why the voltmeter must never be placed in series.
MARKER-PRACTICAL-P6.

Appendix A: full measurement series (R = 250 ohm)

voltage_V	current_mA	resistance_ohm
1.0	4.0	250
1.5	6.0	250
2.0	8.0	250
2.5	10.0	250
3.0	12.0	250
3.5	14.0	250
4.0	16.0	250
4.5	18.0	250
5.0	20.0	250
5.5	22.0	250
6.0	24.0	250
6.5	26.0	250
7.0	28.0	250
7.5	30.0	250
8.0	32.0	250
8.5	34.0	250
9.0	36.0	250
9.5	38.0	250
10.0	40.0	250
10.5	42.0	250
11.0	44.0	250
11.5	46.0	250
12.0	48.0	250